V300 ZEFIRO

A New Sense of Speed

Cab HVAC-VF.CONCEPT SFRS

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1 Introduction

This document is a function specification for the control of the HVAC cab system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:

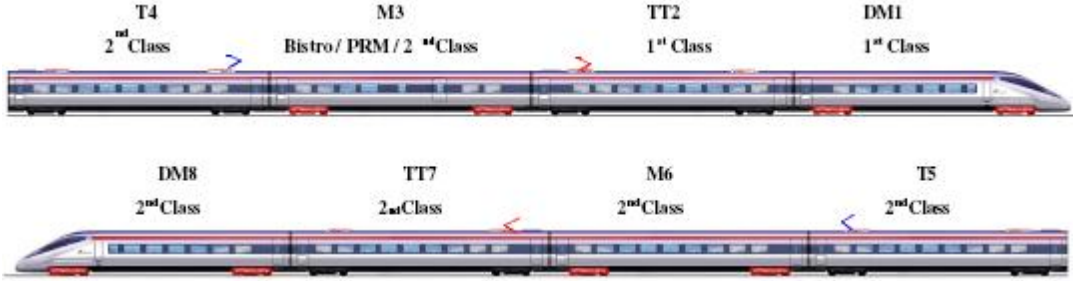


Figure 1 : Train Lay- Out

- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
CU	Control Unit
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
FDS	Fire Detection System
FTP	File Tranfer Protocol
HVAC	Heating Ventilating Air Conditioning
HW	Hardware
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
ID	Identification Number
IDU	Interface Display Unit
IP	Internet Protocoll address
LCP	Local Control Panel
LCU	Local Control Unit
MCM	Master Control Module
N/A	Not Applicable
PC	Portable Computer
RS-232	Recommended Standard 232
SIL	Safety Integrity Level
SW	Software
TCMS	Train Control and Management System
Tim	Internal mean Temperature
TRD	Technical Requirement Document

2 System Overview

The HVAC Control architecture of one car is composed by the following main elements:

- 2 Saloon HVAC modules per coach (1 control unit inside each HVAC saloon module)
- 1 Cab control unit inside each HVAC Cab module. Only in DM1/DM8 cars.

The redundancy of the 2 HVAC roof mounted module is one of the strengths of this system. In case of failure of one controller the other of the car can take over the tasks that the first controller was performing.
Apart of communication buses, there are several discrete signals that are foreseen to receive rapidly important information:

- Tunnel mode pressure transducers.
- Internal fire signal (from fire detection system)

End cars of the train have 2 HVAC saloon units and one HVAC cabin unit. Each of these devices has one control rack inside.
These communications configuration between racks is done as displayed in the figures below:

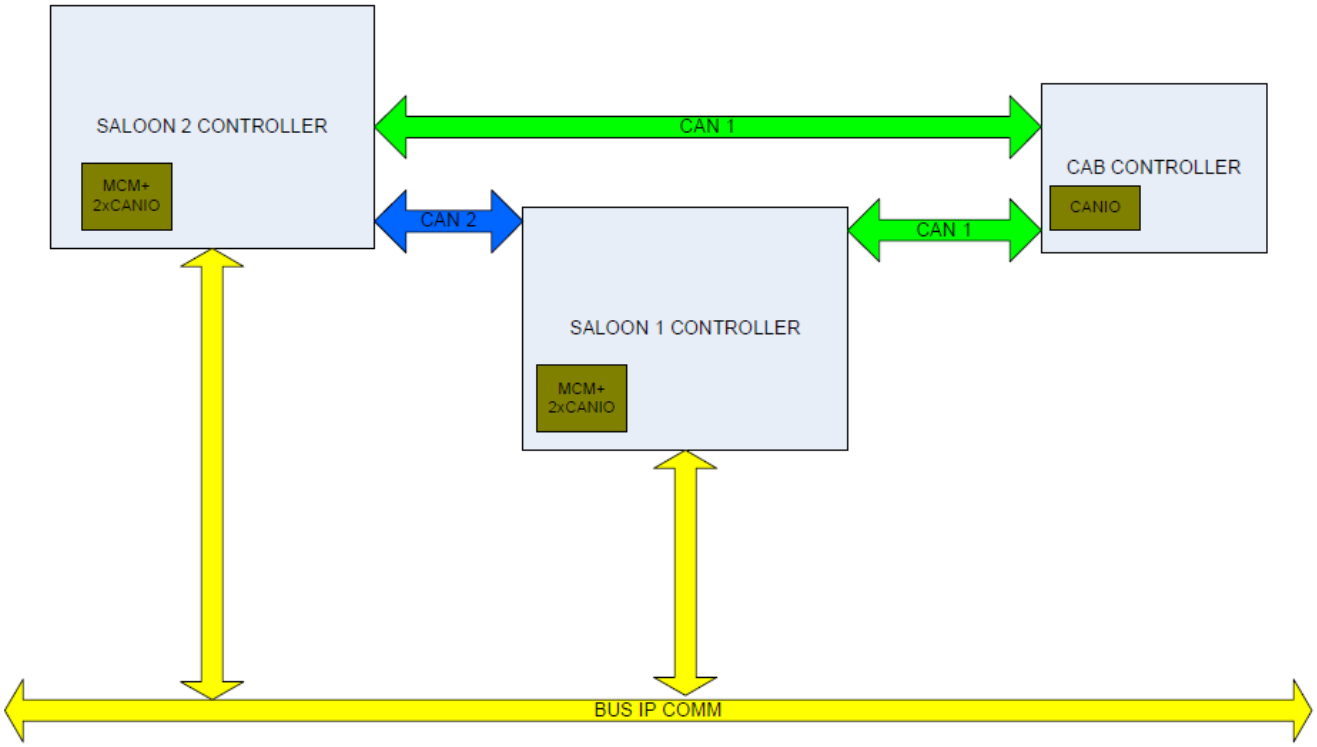


Figure 2 : Communication configuration - *DM1* / *DM8*

3 Functional requirements

3.1 Prioritiy of cab operational modes

3.1.1 Function Design

3.1.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.10
Description	The Safety relevant mode "Fire inside" shall have highest priority.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.	The Safety relevant mode " Fire Inside " shall have highest priority	Derived Requirement	4S_13.-.HVAC		SIL 2

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ID	2F_15.01.Zefiro-Europe.TRS.78
Description	The priority order below the Fire Inside mode shall be: · HVAC OFF mode
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.89	The priority order below the Fire Inside mode shall be: · HVAC OFF mode	Derived Requirement	4S_13.-.HVAC		N/A

3.2 HVAC cab - Fire inside mode

3.2.1 Function Design

In case of FIRE detection (FIRE DETECTION SYSTEM) the HVAC system will be switched OFF by hardwired train line.
Whenever the Fire Detection System is alarmed by FIRE event, the HVAC System of the driver cabin where the fire is detected will be switched off by hardwired train line.
The HVAC of all other coaches where fire is not present will work in full recirculation mode.

3.2.1.1 Normal condition

The HVAC will be switched off by hardwired train line when fire has been detected.
The HVAC Saloon of all other coaches where fire is not present will work in full recirculation mode.
The HVAC Cab where fire is not present will work in full recirculation mode.

3.2.1.2 Failure condition

1. The HVAC Cab cannot enter fire mode when ordered.

3.2.1.3 Failure supervision

1. - A fault will be reported :
   1°) HVAC_FIRE_FAULT signal ;
   2°) HVAC fault code – (i.e Damper fault).
   The TCMS will make a comparison between the status sent by HVAC and the Fire Detection System of the Cab and a diagnostic event shall be displayed to the driver if they differ.
- The driver can switch OFF the HVAC by IDU

3.2.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.18
Description	The cab HVAC shall be switched off by hardwired train line when fire has been detected.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.68	The cab HVAC shall shutdown and close the interface (fresh air, exhaust air) to the exterior part of the train when fire is indicated by Fire Detection System . The information about fire shall be in failsafe logic.	Derived Requirement	4S_13.-.HVAC		SIL 1

2F_15.01.Zefiro-Europe.CONCEPT.130	The cab HVAC shall communicate to TCMS that fire is detected (HVAC_FIRE_MODE signal). If the HVAC can not enter fire mode when ordered a fault shall be reported · HVAC_FIRE_FAULT signal ; · HVAC fault code – (i.e Damper fault).	Derived Requirement	4S_13.-.HVAC		SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.131	The TCMS shall make a comparison between the status sent by HVAC and the Fire Detection System of the same coach and a diagnostic event shall be displayed to the driver if they differ.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

Whenever the Fire Pre – Alarm is set locally by the fire detection system, TCMS will order HVAC_FULL RECIRCULATION_MODE in the Cab where pre-alarm is set (unless no other mode with higher priority are set).

3.3 HVAC cab - Off mode

3.3.1 Function Design

It will be possible to switch OFF all HVAC units in the train by a control on the drivers desk, this is typically used when there is a fire outside the train. It will also be possible for the driver to switch OFF (cut out) individual HVAC units using the IDU.

3.3.1.1 Normal condition

The driver will be able to turn OFF the HVAC Cab system using a control on driver's desk or the IDU.

3.3.1.2 Failure condition

1. The HVAC Cab system cannot enter in HVAC OFF mode by the driver using control on driver's desk
2. The HVAC Cab system cannot enter in HVAC OFF mode ordered by the driver using IDU

3.3.1.3 Failure supervision

1. If the HVAC Cab cannot fulfil the requested OFF mode it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code). The driver will have the possibility to shut down the system by IDU
2. If the HVAC Cab cannot fulfil the requested OFF mode it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code). The driver will have the possibility to shut down the system by control on driver's desk .

3.3.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.15
Description	The driver shall be able to shut down HVAC cab manually via the IDU.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_15.01.Zefiro-Europe.CONCEPT.71	The driver shall be able to turn OFF the HVAC cab manually by IDU : HVAC_OFF_MODE	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

ID	2F_15.01.Zefiro-Europe.TRS.146
Description	The driver shall be able to switch off the cab HVAC.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.138	The driver shall have the possibility to turn off all HVAC units in the train via the drivers desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 1
2F_15.01.Zefiro-Europe.CONCEPT.139	The driver shall have the possibility to turn off all HVAC units in the train via the drivers desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 1

ID	2F_15.01.Zefiro-Europe.TRS.16
Description	The TCMS shall switch off the cab HVAC in all cabs in case of the following states: Normal switch off (for whole train): The TCMS shall switch off the HVAC system in case of the following states (input via IDU or connected with train status): - Wash mode -> from train operational mode - Shut down -> from train operational mode(opening of battery contactor) - HVAC off order per train - HVAC off order from TCMS, if 400 Volt are not available in preconditioning mode/ frost protection mode.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.72	When Wash mode is ordered from train operational mode, TCMS shall order the HVAC units to enter Wash mode, HVAC_WASH_MODE	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.142	If 400 Volt are not available in preconditioning / frost protection mode the HVAC system for that car shall be switched OFF.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

In order to give the HVAC unit time to be shut off in a controlled way TCMS will in the following 2 ways send a pre-warning signal to the HVAC units when the train is about to be shut off for a long time
There will be a pushbutton on the IDU that can be used by the driver/staff to send a pre-warning to the HVAC units. Once the ok is given the driver will lower the pantograph and shut off the train.
A pre-warning will be sent when an automatic exit from Parking mode is requested, the pre-warning will be sent a specified time in advance (10min) before the pantograph is lowered. If not all HVAC units have confirmed that it is OK to shut off an event will be set and the automatic exit from Parking mode will continue.

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.148	The driver shall have the possibility, via a push button on the IDU, to send a pre-waring to the HVAC units that the train is about to be turned off.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_15.01.Zefiro-Europe.CONCEPT.149	When the HVAC unit is OK to be shut off after the pre-waring have been sent by the driver it shall be indicated to TCMS.	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.150	When all HVAC units have confirmed that it is OK to shut off after the pre-waring have been sent by the driver it shall be indicated to the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_15.01.Zefiro-Europe.CONCEPT.408	TCMS shall store an event if the HVAC units not have confirmed that it is OK to be shut down within an adjustable time after the pre-warning command (by driver or during parking mode exit) have been sent. The default value for the adjustable time shall be 180s.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_15.01.Zefiro-Europe.CONCEPT.424	If parking mode is pending to be automatically switched OFF a pre-warning signal shall be sent to the HVAC units	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.4 HVAC cab - Automatic mode

3.4.1 Function Design

Within the HVAC FUNCTIONAL MODE the AUTOMATIC MODE is foreseen.

3.4.1.1 Normal condition

The system will regulate the internal temperature in agreement to comfort requested by standard . The driver will have the possibility to regulate the internal temperature and air velocity by dedicated commands.

3.4.1.2 Failure condition

1. The HVAC cannot regulate at full power the internal comfort with the next conditions. The internal comfort will be more or less degraded depending by external ambient conditions.
a) Failure of the Ventilation system.
b) Failure of the heating system.
c) Failure of cooling system.
2. Complete Failure of all components of both HVAC module.

3.4.1.3 Failure supervision

1. If the HVAC Controller cannot fulfil the requested mode it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code). Then the HVAC Cab unit in degraded mode will operate automatically in order to obtain the best comfort possible with the degraded resources available.
2. If the HVAC Cab Controller cannot fulfil the requested mode it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code). The internal comfort will be provided by conditioned air from Saloon with a dedicated duct system.

3.4.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.24
Description	In automatic mode the cab HVAC system controller shall automatically switch between the modes Cooling, Ventilation and Heating. The interior temperature shall be set depending on the ambient temperature.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.73	Whenever HVAC is working in normal conditions the HVAC shall regulate temperature in agreement to SET point curve obtained by Standards and implemented in HVAC Cab SW	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.152	The HVAC controller shall communicate the mean internal temperature in the cab.	Derived Requirement	4S_13.-.HVAC		N/A

ID	2F_15.01.Zefiro-Europe.TRS.25
Description	When no other mode is requested the cab HVAC system shall automatically operate in normal mode.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.74	The HVAC shall work in HVAC_AUTOMATIC_MODE when no other FUNCTIONAL mode is requested and when no other operational mode with higher priority is	Derived Requirement	4S_13.-.HVAC		N/A

	requested.				
2F_15.01.Zefiro-Europe.CONCEPT.153	The HVAC Controller shall communicate the actual operational mode to TCMS.	Derived Requirement	4S_13.-.HVAC		N/A

3.5 HVAC cab - Communication missing mode

3.5.1 Function Design

In case of communication failure of the IP link of both saloon HVAC units, the HVAC will work in the latest valid operational mode until a mode with higher priority is activated by a hardware input.

3.5.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.40
Description	When the TCMS fails the cab HVAC system shall operate automatically with the latest valid operation mode.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.75	The HVAC cab unit shall work automatically with the latest valid operation mode if TCMS-communication is lost for both saloon HVAC units AND no other mode with higher priority is set.	Derived Requirement	4S_13.-.HVAC		N/A

3.6 HVAC cab - Preconditioning mode

3.6.1 Function Design

This mode can be started only after start-up of the HVAC Cab unit, in case the average internal temperature inside the cab is above set-point temperature. In order to reach the set-point of internal temperature as soon as possible, Fresh air inlets of the HVAC unit will be closed and thus the system will work only with return air.
Conditions to run Preconditioning are:

- ON signal activated.
- Average Internal temperature above set point.
- Ventilation running confirmation by airflow detector.
- “AC voltage available” received from TCMS.
- Cooling system working OK.

This mode ends because of one of the next reasons:

- Tim - 1ºC has been reached in passenger’s area in pre-cooling or Tim + 1ºC in pre-heating
- If there is any indication, that there are passengers onboard (e.g. required platform mode, train moving etc.)
- Maximum pre-conditioning time has elapsed
- Signal “Immediate pre-conditioning Stop” requested by TCMS.

3.6.1.1 Normal condition

The system will operate in Preconditioning mode in order to reach internal target for driver cabin as soon as possible.

3.6.1.2 Failure condition

1. The HVAC cab cannot operate in Preconditioning mode and the temperature inside the cab cannot reach the internal comfort to start the train passenger service.

2. The HVAC Automatic mode will not start after preconditioning phase

3.6.1.3 Failure supervision

1. If the HVAC Controller cannot fulfil the requested Preconditioning mode it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code).
2. If the HVAC Controller cannot fulfil the requested automatic mode after preconditioning phase, it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code).

3.6.2 Requirements

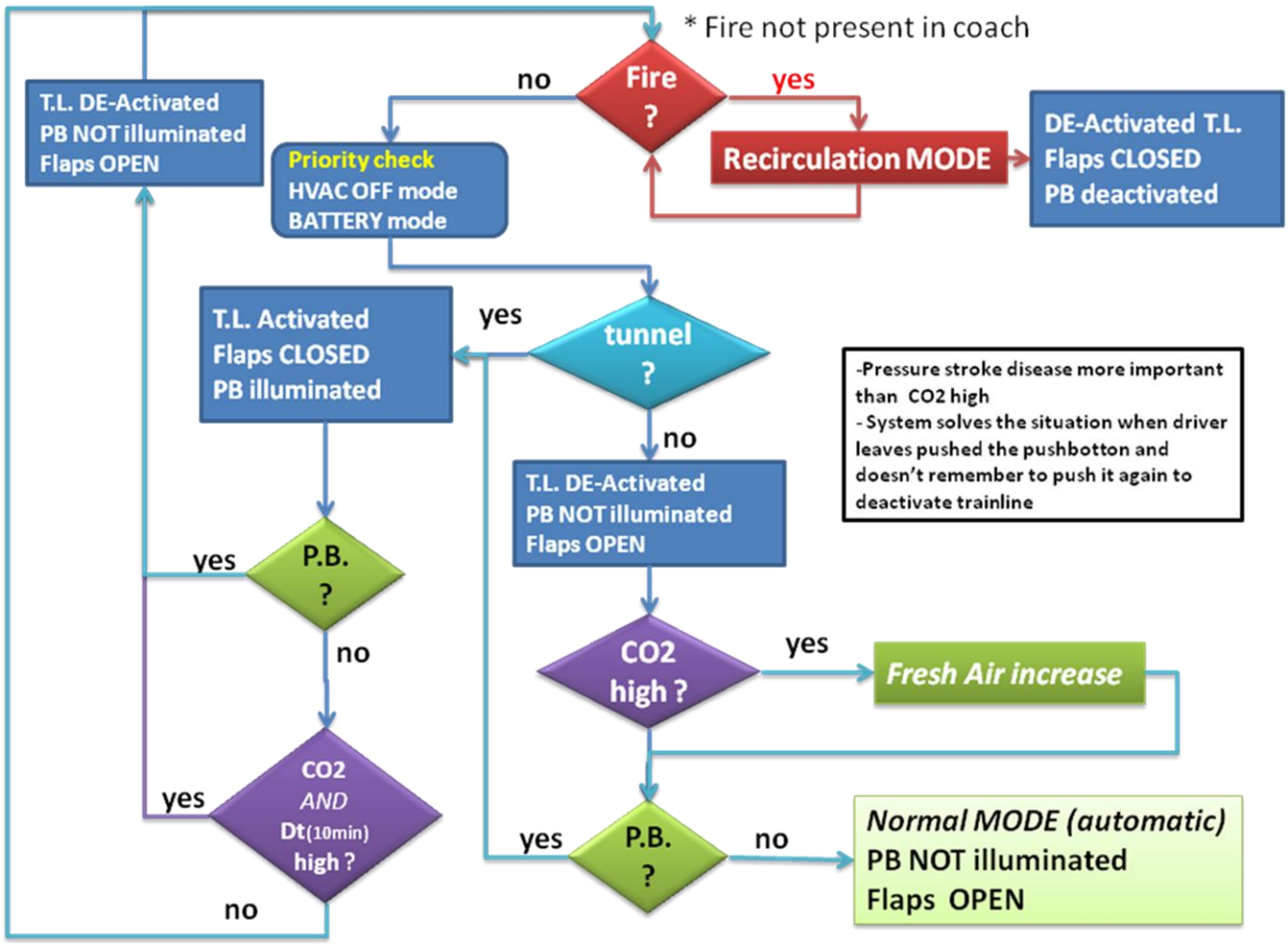
ID	2F_15.01.Zefiro-Europe.TRS.44
Description	The preconditioning mode of HVAC cab unit shall be activated and deactivated in parallel with preconditioning mode of HVAC saloon unit.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.76	TCMS shall send to HVAC Cab Controller the HVAC_PRECONDITIONING_MODE signal after start-up of the train and when line voltage is available	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.01.Zefiro-Europe.CONCEPT.166	Whenever the driver requests to conclude the HVAC_PRECONDITIONING_MODE phase by IDU the TCMS shall set the HVAC Cab Controller in HVAC_AUTOMATIC_MODE if no other functional mode are requested.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.01.Zefiro-Europe.CONCEPT.168	The HVAC shall shift from HVAC_PRECONDITIONING_MODE to HVAC_AUTOMATIC_MODE (if no other functional mode are requested) if the internal targets in terms of comfort are achieved.	Derived Requirement	4S_13.-.HVAC		N/A

3.7 HVAC cab - Tunnel mode

3.7.1 Function Design

The pressure detection sensors will measure the pressure difference between the internal and external part of the train. The value measured out by the sensors will permit the HVAC LCU to foresee the pressure waves that can appear when entering a tunnel.



Different Scenarios :

-Tunnelmode is OFF: Driver activates PB, TCMS orders HVAC to tunnelmode, HVAC activates DO and activates tunnelmode TL and TCMS illuminates the PB

-Tunnelmode is OFF: HVAC activates tunnelmode due to pressure transducers, TCMS recieves status of tunnelmode TL and illuminates the PB

-Tunnelmode is ON, activated by cab HVAC: Driver presses tunnelmode PB, TCMS orders HVAC to turn tunnelmode OFF, HVAC removes DO and deactives tunnelmode TL and TCMS switch OFF the PB lamp

-Tunnelmode is ON, activated by driver: Driver presses tunnelmode PB, TCMS orders HVAC to remove tunnelmode, HVAC removes DO and deactives tunnelmode TL and TCMS switch OFF the PB lamp

In case of Fire:
During fire HVAC units in cars where fire is not present will be set to recirculation mode, ordered by TCMS. The tunnelmode PB on drivers desk is NOT active, TCMS will not set DO. HVAC units FLAPS shall be closed individually by each HVAC controller and Tunnelmode TL will be deactivated, PB on drivers desk will not be illuminated and will be deactivated.

It will be also possible for the driver to request Tunnelmode using a push button on the drivers desk.

When Tunnel mode is set, either by the pressure transducers or by the driver, this is indicated by illuminating the push button on the drivers desk.

3.7.1.1 Normal condition

The system will operate in Tunnel mode when requested by Driver or by pressure transducers in order to avoid the pressure wave disease.

3.7.1.2 Failure condition

1. The HVAC cannot operate in Tunnel mode when requested by pressure transducers.
2. The HVAC cannot operate in Tunnel mode when requested by Driver via Tunnel mode Push Bottom.
3. The dampers cannot open again after HVAC Tunnel mode.

3.7.1.3 Failure supervision

1. If the HVAC Controller cannot fulfil the requested Tunnel mode it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code).
2. If the HVAC Controller cannot fulfil the requested Tunnel mode it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code).
3. If the HVAC Controller cannot fulfil the requested Tunnel mode it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code).

3.7.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.76
Description	When signal from differential pressure switch is activated and if no other mode with higher priority is set, all HVAC systems shall operate in tunnel mode. The driver shall also be able to activate and de-activate tunnel mode manually.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.77	When signal from differential pressure transducers is activated output “HVAC tunnel mode” shall be processed by HVAC cab unit and the HVAC cab unit shall enter Tunnel mode to close HVAC Cab pneumatic dampers. When tunnel mode no longer is indicated by the differential pressure transducers Tunnel mode shall be left and the HVAC Cab pneumatic dampers shall be opened.	Derived Requirement	4S_13.-.HVAC		SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.170	When Tunnelmode is indicated by the HVAC cab unit output, the tunnel mode Trainline“HVAC tunnel mode” shall be set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.171	When “Tunnelmode” is indicated on the HVAC cab unit input the HVAC cab unit shall enter Tunnel mode to close HVAC Cab pneumatic dampers.	Derived Requirement	4S_13.-.HVAC		SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.172	The driver in the activated cab shall have the possibility to activate and deactivate “HVAC tunnel mode” for all HVAC units in the train via the drivers desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.411	The driver shall have the possibility to activate and de-actvate “HVAC tunnel mode” in the train via the drivers desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_15.01.Zefiro-Europe.CONCEPT.	TCMS shall order all HVAC units to enter “HVAC tunnel mode” when ordered by the driver via the	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

173	drivers desk.				
2F_15.01.Zefiro-Europe.CONCEPT.425	When the HVAC cab unit is ordered to Tunnel mode by TCMS, the output “HVAC tunnel mode” shall be set by HVAC cab unit. When tunnel mode no longer ordered by TCMS Tunnel mode shall be left and the HVAC Cab pneumatic dampers shall be opened.	Derived Requirement	4S_13.-.HVAC		SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.174	TCMS shall compare status from all HVAC units when “HVAC tunnel mode” is requested and a diagnostic event shall be displayed to the driver if they differ.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.412	When “HVAC Tunnelmode” TL” is indicated by any of the HVAC cab units the Tunnelmode pushbutton on the drivers desk in the activated cab shall be illuminated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.417	When “HVAC Tunnelmode” TL” is indicated by TCMS the Tunnelmode pushbutton on the drivers desk in the activated cab shall be illuminated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.413	Status of “HVAC Tunnelmode” TL shall be sent to TCMS.	Derived Requirement	4S_13.-.HVAC		SIL 0

ID	2F_15.01.Zefiro-Europe.TRS.74
Description	The air inlet and outlet flaps shall be possible to close when running in tunnel mode.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.78	Whenever the HVAC is in HVAC_TUNNEL_MODE the HVAC unit shall close all the air inlet sections and all the flaps in order to avoide the pressure wave discomfort for driver	Derived Requirement	4S_13.-.HVAC		N/A

3.8 HVAC Cab - Recirculation mode

3.8.1 Function Design

Whenever a fire is detected, the TCMS will visualize the Fire event on the IDU and by a relevant visual signal on the driver’s desk.
Whenever a fire is detected the HVAC of all other coaches where fire is not present will work in full recirculation mode, this is covered in the SFRS Fire Detection and Extinction ref [3].
If the HVAC units are in Tunnelmode and fire is detected, the HVAC units in the cars where fire is not present will enter recirculation mode and the Tunnel mode push button on the drivers desk will not be illuminated and not possible to use.

3.8.2 Normal condition

Recirculation mode is entered when requested.

3.8.2.1 Failure condition

1. Recirculation mode cannot be entered.

3.8.2.2 Failure supervision

1 If the HVAC Controller cannot fulfil the requested mode it will be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code).
The driver can use the HVAC OFF selector on the drivers desk to turn all HVAC units OFF.

3.8.3 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.151
Description	When fire has been detected in one car the HVAC units in the others cars shall be set in recirculation mode.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.415	When recirculation mode is activated it shall not be possible to activate Tunnel mode by the driver push button. The Tunnelmode PB shall be ignored.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_15.01.Zefiro-Europe.CONCEPT.416	Reciculation mode shall have higher priority than Tunnelmode. When reciculation mode is requested and the HVAC unit is in Tunnelmode, recirculation mode shall be entered and output “HVAC tunnel mode” shall be reset	Derived Requirement	4S_13.-.HVAC		SIL 0

3.9 HVAC Cab - Compressor start

3.9.1 Function Design

In order to avoid that different cooling compressors of the HVAC Equipments starts simultaneously, overloading the train Auxiliary Converters, compressor startup of each saloon and cab HVAC unit of the train set will be managed by a request-permission sequence under control of the TCMS

3.9.1.1 Normal condition

HVAC controller shall send to TCMS a start request when it wants to start it’s compressors.

3.9.1.2 Failure condition

1. The compressor of the HVAC Cab unit cannot start and it will not be possible to have cooling power for driver during summer.

3.9.1.3 Failure supervision

1. The HVAC Cab System will open the BY-PASS Damper in order to receive treated air by HVAC saloon

3.9.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.149
Description	The HVAC compressors shall be started sequentially to ensure that the start-up load is not too high.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.211	HVAC controller shall send a start request when it want to start it’s compressors : HVAC_COMPRESSOR_START signal	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.212	When the startup request from a HVAC controller compressor is active TCMS shall send a start order to that HVAC controller compressor when no other HVAC controller compressor startup request are active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.01.Zefiro-Europe.CONCEPT.	The HVAC control compressor shall start after receiving the startup permission from TCMS	Derived Requirement	4S_13.-.HVAC		N/A

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2F_15.01.Zefiro-Europe.CONCEPT.214	The HVAC Controller shall communicate to TCMS that the compressor is activated and remove the start request	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.215	TCMS shall remove the start order after a specified time or when the startup request is removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.10 HVAC Cab - Maintenance mode

3.10.1 Function Design

The Ethernet Network for SW download and to perform all the service interface activities will be used.

According to reference [1] the TRD for HVAC, chapter 4.9.6.2.2 : *" The system will provide over the IP network interface to TCMS a service interface for remote connection for central on-board use as well as for direct local connection when disconnected from the TCMS network. No second Ethernet connection should be foreseen.The system shall provide additionally via the IP network interface the functions for Software download, configuration download, access to system local diagnostics, access to system local parameters. "*

The PC will be connected to suitable switch on the network to communicate with the HVAC unit by the use of FTP in agreement with [2] Network specific requirements.
This mode will be activated by request of external device (e.g. PC) when the train is in the depot..

3.10.1.1 Normal condition

The system will operate in Maintenance mode when a PC is linked to Ethernet connection and when maintenance activities are necessary during train depot mode.

3.10.1.2 Failure condition

1. The HVAC cannot operate in Maintenance mode and no connection between PC and HVAC controller can be provided .
2. The SW cannot be uploaded in HVAC Controller.

3.10.1.3 Failure supervision

1. There is the possibility to link the PC directly to HVAC controller by RS 232 serial port inside HVAC module roof mounted.
2. The SW can be uploaded in HVAC Controller by RS 232 serial port present inside the module roof mounted.

3.10.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.154
Description	It shall be possible to set the HVAC in test mode.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.219	It shall be possible to set the HVAC controller in testmode in order to perform trouble shooting activities. It shall be possible to set the HVAC cab in test	Derived Requirement	4S_13.-.HVAC		SIL 0

	mode by PC through the IP network interface to TCMS for remote connection for central on-board use as well as for direct local connection when disconnected from the TCMS network.				

3.11 HVAC cab - Interface with HVAC saloon

3.11.1 Function Design

The HVAC Saloon shall be interfaced with HVAC cab in order to allow exchange data

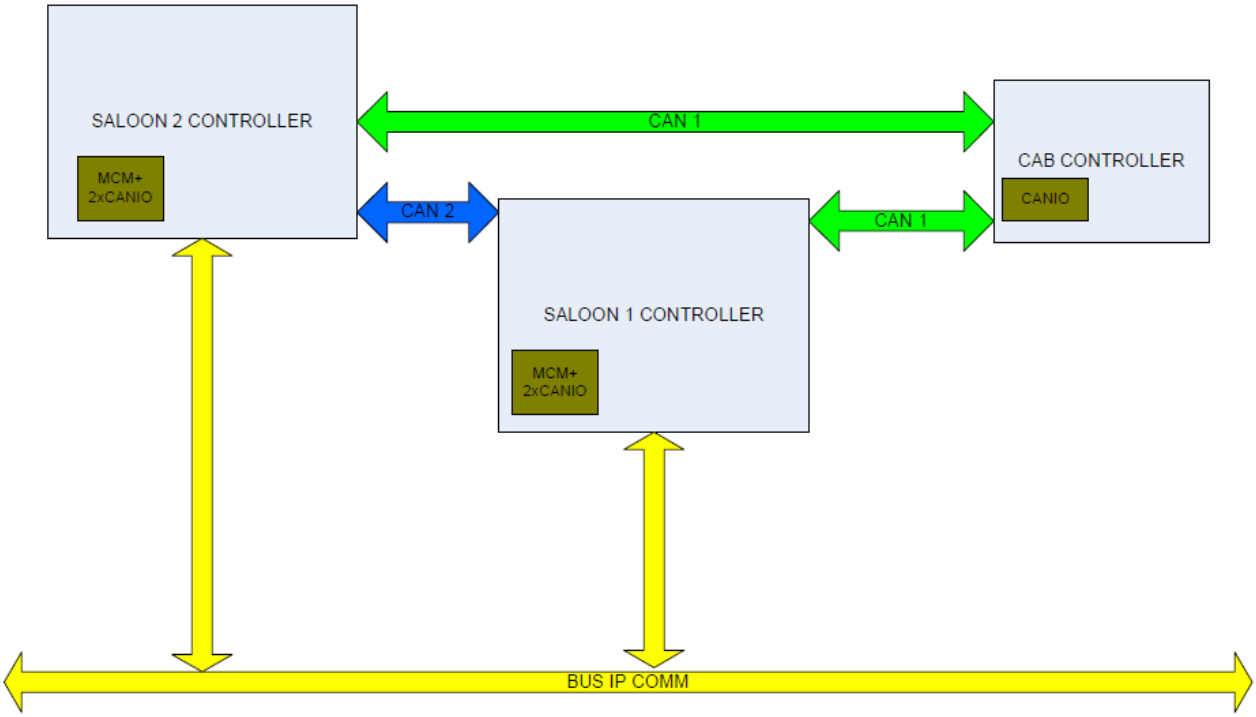


Figure 7 : Interface with HVAC saloon - *DM1 / DM8*

3.11.1.1 Normal condition

Both HVAC controller Saloon will communicate with HVAC controller CAB in order to exchange relevant data with Train Computer.
The HVAC Cab unit will provide internal comfort automatically.

3.11.1.2 Failure condition

1. The communication of both HVAC controller Saloon with HVAC controller Cab is not possible and there will not be the possibility to exchange data with TCMS.
2. The HVAC unit roof mounted will not provide internal comfort for driver for HVAC cab module fault

3.11.1.3 Failure supervision

1. In case of communication failure of both HVAC Saloon controller the HVAC Cab will work in the latest valid HVAC Cab functional mode if no other modes with higher priority are set.
2. HVAC Saloon will provide nominal level of comfort in the cab in case of fault of components inside the HVAC cab unit. In this case the treated air will be provided directly by HVAC saloon ducting through a dedicated ducting by pass .

3.11.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.45
Description	The HVAC saloon shall be interfaced with the HVAC cab.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.79	Both the HVAC Controller of Saloon shall have the possibility to communicate all the relevant data and HVAC saloon status to the HVAC Controller of the HVAC Driver Cabin Unit	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.222	The HVAC Controller of saloon shall have the possibility to receive all the relevant data and HVAC Driving Cab status	Derived Requirement	4S_13.-.HVAC		N/A

ID	2F_15.01.Zefiro-Europe.TRS.42
Description	The nominal level of comfort shall then be provided in case of fault to the cab HVAC.
Safety level	N/A

The nominal level of comfort shall be provided from the HVAC saloon units in case of fault to the cab HVAC
Whenever an HVAC cab unit fault occur and no air conditioning can be provided directly by HVAC cab unit, it will provided by the HVAC saloon units.

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.223	HVAC Saloon shall provide nominal level of comfort in the cab in case of fault in the HVAC cab.	Derived Requirement	4S_13.-.HVAC		N/A

3.12 HVAC cab - Control of temperature

3.12.1 Function Design

Under nominal operative conditions, the temperature of the comfort envelope will be automatically controlled in dependence of the set point (Tic) temperature selection of the driver by the local control panel on the driver desk.

3.12.1.1 Normal condition

The driver will have the possibility to change the temperature set point in order to maximize the comfort.

3.12.1.2 Failure condition

1. The system cannot change the temperature set point as requested by the driver.

3.12.1.3 Failure supervision

1. In case of communication failure, the temperature will remain the set point comfort curve requested by standard EN 14813 related to the exterior temperature.

3.12.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.50
Description	The cab room temperature shall be adjusted from the comfort curve.
Safety level	N/A

Under nominal operative conditions, the temperature of the Cab will be automatically controlled in dependence of the set point (Tic) temperature selection of the driver by the local control panel on the driver desk.

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.229	The selected interior temperature setpoint step shall be sent to TCMS.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_15.01.Zefiro-Europe.CONCEPT.418	The driver´s cab shall have a selection switch in order to set offset from the nominal set-point value of interior temperature. (ΔTic) There shall be 3 offsets; -2 K; 0; +2 K	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A

ID	2F_15.01.Zefiro-Europe.TRS.48
Description	The driver shall be able to adjust the cab room temperature offset in the activated cab.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.231	Depending on the temperature selection switch offset (ΔTic) and the mean exterior temperature (Tem), the interior temperature setting (Tic) shall be controlled according to the following equation: Tic = 20 + ΔTic + 0,4 (Tem – 20) for Tem ≥ + 20 °C; Tic = 20 + ΔTic for Tem < + 20 °C; The interior temperature setting (Tic) in the activated cab shall be sent to the HVAC cab unit.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.01.Zefiro-Europe.CONCEPT.232	The HVAC cab controller shall communicate the reading sensors temperature to the TCMS	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.233	The TCMS shall receive the internal temperature by HVAC cab controller and shall show the temperature by IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

3.13 HVAC cab - Control of airflow

3.13.1 Function Design

An adjustable override will be provided (by the local control panel on the driver desk or by IDU) to ensure individual driver's comfort. It will be possible to select from a "high" (level 4) or a "low" (level 1) fan speed setting. Selection of "high" speed will allow the air conditioning installation to continue to operate at its maximum capacity.

3.13.1.1 Normal condition

The driver will have the possibility to change the air flow in order to maximize the comfort.

3.13.1.2 Failure condition

1. The system cannot change the air flow as requested by the driver.

3.13.1.3 Failure supervision

1. In case of communication failure, the air flow will remain set in position “Auto”.

3.13.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.52
Description	The driver shall be able to change the air flow level in the cab.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.83	An automatic fan speed control (by regulation damper inside HVAC module) shall be provided. A manual override shall be provided (by the local control panel on the driver desk) to ensure individual driver’s comfort. It shall be possible to select from a "high" (level 4) or a "low" (level 1) fan speed setting. Selection of "high" speed shall allow the air conditioning installation to continue to operate at its maximum capacity.	Derived Requirement	4S_13.-.HVAC	HW HMI	N/A

ID	2F_15.01.Zefiro-Europe.TRS.53
Description	The air flow level positions shall be Stop, Auto, 1, 2, 3 and 4.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.84	The local control panel on the driver desk shall have the following air flow level positions - Stop, - Auto, - 1 -> LOW air flow - 2 - 3 - 4 -> HIGH air flow	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_15.01.Zefiro-Europe.CONCEPT.235	The airflow requested by the driver using the air flow control on the drivers desk shall be sent to the HVAC cab unit.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.01.Zefiro-Europe.CONCEPT.420	The driver shall have the possibility to change the level of cab airflow via the drivers desk with the following air flow level positions - Stop, - Auto, - 1 -> LOW air flow - 2 - 3 - 4 -> HIGH air flow	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_15.01.Zefiro-Europe.CONCEPT.421	The HVAC shall regulate the airflow level according to the selected position by the driver - Stop, - Auto, - 1 -> LOW air flow - 2 - 3	Derived Requirement	4S_13.-.HVAC		N/A

	- 4 -> HIGH air flow				

ID	2F_15.01.Zefiro-Europe.TRS.54
Description	The air flow level and temperature set point in a non active cab shall be set to Auto regardless of the switch position.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.85	TCMS shall send to HVAC Cab controller the information regarding the status of the Cab : “Activated” ; Not activated”	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.01.Zefiro-Europe.CONCEPT.236	The air flow level in the non active cab shall be set automatically in “Auto” mode regardless of the switch position local control panel	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.422	The HVAC status of the NON activated cab shall be RECIRCULATION MODE	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.423	The temperature Set Point shall be the selected interior temperature setpoint of the activated CAB	Derived Requirement	4S_13.-.HVAC		N/A

3.14 HVAC cab - Condition data

3.14.1 Function Design

The status of the HVAC system will be indicated on the IDU.

3.14.1.1 Normal condition

The HVAC Cab controller shall send to TCMS the Status of the system

3.14.1.2 Failure condition

1. The HVAC Cab controller cannot communicate the status of the system

3.14.1.3 Failure supervision

1. In case of communication failure, a relevant error message will be shown to the driver and the system will work in “Auto” mode.

3.14.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.64
Description	The status of the HVAC system shall be indicated on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.86	The HVAC cab Controller shall send to TCMS the complete diagnostic of the driver cab HVAC systems	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.238	The TCMS shall receive all the Diagnostic Status of driver cab HVAC and shall communicate all datas available by IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_15.01.Zefiro-	The operational modes "On", "Off", "Cut out" and	Derived	4S_09.03.05.-.TCMS	IDU	N/A

Europe.CONCEPT.154	"Faulty" shall be displayed to the driver.	Requirement	Software		

3.15 HVAC cab - Fault supervision

3.15.1 Function Design

3.15.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.60
Description	The HVAC shall supervise itself and a fault indication shall be set in case of component faults.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.87	The HVAC Controller shall supervise itself and a fault indication shall be set in case of component faults.	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.155	If the HVAC Controller cannot fulfil the requested mode it shall be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code).	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.156	TCMS shall store an event when the HVAC_GENERAL_FAULT, HVAC fault code is received from the HVAC Controller	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_15.01.Zefiro-Europe.TRS.155
Description	Complete diagnostic of the cab HVAC system, including predictive diagnostic functions and CBM, shall be fully implemented in each cab HVAC unit controller. The data shall be transmitted to the TCMS via the saloon HVAC bus interface.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.240	The HVAC Controller shall send to TCMS the complete diagnostic of the cab HVAC system	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.01.Zefiro-Europe.CONCEPT.241	The TCMS shall receive all the Diagnostic Status of each HVAC cab and store it in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.16 Events

3.16.1 Function Design

3.16.2 Requirements

ID	2F_15.01.Zefiro-Europe.TRS.143
Description	Diagnostics from the supplier shall be stored in the train diagnostic system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_15.01.Zefiro-Europe.CONCEPT.118	Diagnostics from the supplier shall be stored in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_15.01.Zefiro-Europe.CONCEPT.242	HVAC tunnel mode	L	2F_15.01.Zefiro-Europe.CONCEPT.171
2F_15.01.Zefiro-Europe.CONCEPT.28	Fire detected	L	2F_15.01.Zefiro-Europe.CONCEPT.67 2F_15.01.Zefiro-Europe.CONCEPT.138

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_15.01.Zefiro-Europe.CONCEPT.29	HVAC tunnel mode	L	2F_15.01.Zefiro-Europe.CONCEPT.77 2F_15.01.Zefiro-Europe.CONCEPT.413

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_15.01.Zefiro-Europe.CONCEPT.30	Cab temperature setpoint switch	A	2F_15.01.Zefiro-Europe.CONCEPT.229

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_15.01.Zefiro-Europe.CONCEPT.31	HVAC tunnel mode PB	L	2F_15.01.Zefiro-Europe.CONCEPT.412 2F_15.01.Zefiro-Europe.CONCEPT.416

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_15.01.Zefiro-Europe.CONCEPT.32	HVAC_FIRE_FAULT	L	2F_15.01.Zefiro-Europe.CONCEPT.131
2F_15.01.Zefiro-Europe.CONCEPT.243	HVAC_GENERAL_FAULT	L	2F_15.01.Zefiro-Europe.CONCEPT.87 2F_15.01.Zefiro-Europe.CONCEPT.155
2F_15.01.Zefiro-Europe.CONCEPT.244	HVAC fault code	A	2F_15.01.Zefiro-Europe.CONCEPT.87 2F_15.01.Zefiro-Europe.CONCEPT.131 2F_15.01.Zefiro-Europe.CONCEPT.155
2F_15.01.Zefiro-Europe.CONCEPT.245	HVAC_OPERATIONAL_MODE	A	2F_15.01.Zefiro-Europe.CONCEPT.153 2F_15.01.Zefiro-Europe.CONCEPT.174 2F_15.01.Zefiro-Europe.CONCEPT.222
2F_15.01.Zefiro-Europe.CONCEPT.246	HVAC_MEAN_INT_TEMP	A	2F_15.01.Zefiro-Europe.CONCEPT.152
2F_15.01.Zefiro-Europe.CONCEPT.247	HVAC OK to shut OFF	L	2F_15.01.Zefiro-Europe.CONCEPT.149
2F_15.01.Zefiro-Europe.CONCEPT.	HVAC fire detected	L	2F_15.01.Zefiro-Europe.CONCEPT.131

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2F_15.01.Zefiro-Europe.CONCEPT.250	HVAC_COMPRESSOR_START request	L	2F_15.01.Zefiro-Europe.CONCEPT.211
2F_15.01.Zefiro-Europe.CONCEPT.414	Status “HVAC Tunnelmode” TL	L	

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_15.01.Zefiro-Europe.CONCEPT.33	Operation Mode request	A	2F_15.01.Zefiro-Europe.CONCEPT.71 2F_15.01.Zefiro-Europe.CONCEPT.72 2F_15.01.Zefiro-Europe.CONCEPT.76 2F_15.01.Zefiro-Europe.CONCEPT.78
2F_15.01.Zefiro-Europe.CONCEPT.253	HVAC_COMPRESSOR_START order	L	2F_15.01.Zefiro-Europe.CONCEPT.212
2F_15.01.Zefiro-Europe.CONCEPT.256	Interior temperature setpoint	A	2F_15.01.Zefiro-Europe.CONCEPT.231
2F_15.01.Zefiro-Europe.CONCEPT.257	Local cab active	L	2F_15.01.Zefiro-Europe.CONCEPT.85
2F_15.01.Zefiro-Europe.CONCEPT.258	Fan speed setting selector	A	2F_15.01.Zefiro-Europe.CONCEPT.83 2F_15.01.Zefiro-Europe.CONCEPT.235
2F_15.01.Zefiro-Europe.CONCEPT.259	Shutdown Prewarning	L	2F_15.01.Zefiro-Europe.CONCEPT.148 2F_15.01.Zefiro-Europe.CONCEPT.424

5 References

[1] EC01P006696B TRD Passenger saloon and driving cab HVAC system
[2] 3EGH000035-1410 Network specific requirements
[3] EC01P008884B SFRS Fire Detection and Extinction